

TAMPA INTL, FL, USA

WMO: 722110

Lat: 27.962N

Lon: 82.540W

Elev: 19

StdP: 14.69

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 12842

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.8	43.7	23.1	17.4	47.6	27.9	21.8	50.2	20.9	59.8	18.8	59.5	6.6	10	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	12.9	92.5	76.8	91.3	77.0	90.3	76.9	80.4	87.9	79.9	87.5	79.1	86.8	8.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.4	147.5	85.0	77.5	143.0	84.2	76.9	140.2	83.8	44.0	88.0	43.1	87.4	42.5	86.7	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation	n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	(p)
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(4) 18.1	16.0	13.4	DB	32.9	95.4	4.0	1.3	30.0	96.3	27.7	97.1	25.4	97.8	22.5	98.8	
(5)			WB	28.2	82.2	3.8	1.5	25.4	83.3	23.2	84.1	21.0	84.9	18.2	86.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.9	60.9	64.2	67.9	73.4	79.0	82.2	83.0	83.3	82.0	76.7	69.1	64.6
	DBStd	9.59	8.21	7.65	6.69	4.86	3.48	2.58	2.17	2.13	2.49	5.14	6.31	8.03
	HDD50	23	14	4	0	0	0	0	0	0	0	0	0	5
	HDD65	481	176	99	51	7	0	0	0	0	0	4	34	110
	CDD50	8752	353	401	556	701	898	967	1023	1033	959	828	574	459
	CDD65	3733	50	76	142	257	433	517	558	568	509	367	158	98
	CDH74	34112	163	295	728	1737	3964	5434	6157	6297	5138	2960	851	388
	CDH80	11913	8	29	95	404	1407	2129	2422	2516	1892	856	124	31
Wind	WSAvg	6.7	6.9	7.1	7.7	7.7	7.0	6.5	5.6	5.6	6.1	6.6	6.6	6.5
Precipitation	PrecAvg	46.7	2.3	2.9	3.0	1.7	2.8	6.3	7.4	8.2	6.3	2.1	1.5	2.3
	PrecMax	67.7	6.3	10.8	12.0	10.7	17.6	18.7	20.6	16.5	15.3	6.2	6.1	15.6
	PrecMin	29.8	0.1	0.3	0.0	0.0	0.0	1.5	1.7	2.8	0.8	0.0	0.0	0.1
	PrecStd	10.0	1.5	2.2	2.6	1.7	3.0	3.5	3.3	3.1	3.4	1.5	1.4	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.1	83.4	84.9	88.3	93.1	94.1	93.4	93.5	92.9	90.4	85.6	83.0
		MCWB	69.8	70.8	68.9	71.0	73.3	76.8	77.2	77.9	76.4	75.8	73.1	71.6
	2%	DB	78.0	80.2	82.6	86.1	90.3	91.9	91.6	91.9	91.2	88.3	83.3	80.5
		MCWB	68.0	69.2	69.1	71.0	73.4	76.7	77.7	78.1	76.2	75.1	71.7	70.5
	5%	DB	75.6	77.7	80.6	84.3	88.5	90.4	90.4	90.7	89.8	86.6	81.2	78.4
		MCWB	67.2	68.0	68.7	70.6	73.1	76.7	77.8	78.2	76.4	74.3	70.5	69.6
	10%	DB	73.0	75.2	78.4	82.4	86.8	88.9	89.0	89.4	88.4	84.8	79.1	75.9
		MCWB	65.9	66.7	67.8	69.8	72.8	76.6	77.8	78.2	76.4	73.5	69.2	68.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.7	73.1	75.1	76.2	78.3	80.6	81.4	82.1	80.7	79.3	76.3	74.4
		MCDB	77.5	79.7	80.3	83.3	86.7	88.7	88.0	89.4	87.4	85.2	81.5	79.0
	2%	WB	70.7	71.4	73.2	74.8	77.1	79.6	80.3	80.5	79.7	78.1	74.8	73.0
		MCDB	75.0	76.9	78.3	81.7	85.2	87.8	87.6	88.1	86.9	84.2	79.9	77.2
	5%	WB	68.9	70.1	71.7	73.4	76.1	78.6	79.5	79.8	78.7	77.1	73.2	71.5
		MCDB	73.2	75.3	76.6	80.5	84.0	86.7	87.2	87.7	85.8	83.2	78.2	75.7
	10%	WB	67.2	68.7	70.2	72.3	75.2	78.0	78.6	79.0	78.0	76.1	71.5	70.1
		MCDB	71.7	73.5	75.3	79.2	82.9	85.9	86.2	86.8	84.9	81.9	76.5	74.6
Mean Daily Temperature Range	5% DB	MDBR	17.2	16.9	16.9	16.5	16.0	13.6	12.7	12.9	13.2	15.0	16.7	16.2
		MCDBR	16.7	16.8	17.3	17.3	17.3	15.2	13.9	14.2	14.7	14.8	16.2	15.1
		MCWBR	9.0	8.4	7.8	7.1	6.3	5.8	5.6	5.4	5.1	5.6	7.6	7.5
	5% WB	MCDBR	14.2	14.3	13.8	14.8	14.5	13.2	12.6	12.8	12.5	12.4	13.4	13.0
		MCWBR	9.4	8.5	8.8	8.0	6.5	5.9	6.0	5.9	5.6	5.8	8.2	8.4
Clear-Sky Solar Irradiance	taub	0.343	0.355	0.368	0.392	0.425	0.450	0.474	0.455	0.422	0.390	0.354	0.342	
	taud	2.520	2.492	2.444	2.343	2.309	2.316	2.254	2.321	2.383	2.456	2.501	2.548	
	Ebn at Noon	286	291	293	287	276	267	261	266	272	276	279	281	
	Edh at Noon	29	32	36	40	42	41	44	41	37	33	29	27	
All-Sky Solar Radiation	RadAvg	1121	1353	1713	2055	2178	1964	1857	1790	1630	1504	1225	1037	
	RadStd	80	88	141	102	120	157	113	112	106	94	64	85	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.95	+1.93	N/A	+0.36	N/A	-0.45	-9	-92	+333	+225
(47)	Regional (38 neighbors)	+0.76	N/A	N/A	N/A	N/A	N/A	-7	-85	+295	+209

Nomenclature: See separate page